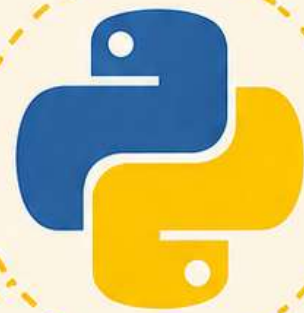


04
PHASE

Python for Data Analyst

Code. Analyze. Automate.
Build **Real Impact**.



Data Handling
with Pandas



Data Analysis
& Visualization



Automation
& Workflows



Build Scalable
Solutions



Transform Data. Write Python. Drive Decisions.

Table of Contents

14 Units • 112 Topics • A Complete Analytics Journey

Unit 1 Python Fundamentals for Analytics.....10 - 21

- 1.1 Variables, Constants, and PEP 8 Naming Conventions
- 1.2 Primitive Data Types: String, Integer, Float, Boolean
- 1.3 String Manipulation: f-strings, .strip(), .split(), .replace()
- 1.4 Type Casting and Checking (type(), isinstance())
- 1.5 Mathematical Operators & Order of Operations
- 1.6 Logical Operators (and, or, not) and Membership (in)
- 1.7 Conditional Logic: if, elif, else
- 1.8 Comparison Operators for Data Filtering

Unit 2 Data Structures & Iteration (The Logic)22 - 37

- 2.1 Lists: Slicing, appending, and list methods
- 2.2 Dictionaries: Key-value pairs for JSON-like data
- 2.3 Sets & Tuples: Unique values and immutability
- 2.4 For Loops: Iterating through lists and dictionaries
- 2.5 While Loops: Logic-based iteration
- 2.6 List Comprehensions: Writing "Pythonic" one-liners for data transformation
- 2.7 Dictionary Comprehensions: Transforming mapping data
- 2.8 Nested structures: Lists of dictionaries (the standard API format)

Unit 3 Functions & Error Handling.....38 - 57

- 3.1 Defining Functions: def, arguments, and return
- 3.2 Positional vs. Keyword arguments
- 3.3 Lambda Functions: Anonymous functions for quick data mapping
- 3.4 Scope: Local vs. Global variables
- 3.5 Error Handling: try, except, finally to prevent script crashes
- 3.6 Importing modules (import math, import os)
- 3.7 Understanding `__name__ == "__main__"`
- 3.8 Documentation: Writing Docstrings for collaborative code

Unit 4 NumPy — The Foundation of Data.....58 - 77

- 4.1 The ndarray: Why it's faster than Python lists
- 4.2 Array creation: np.array(), np.zeros(), np.arange(), np.linspace()
- 4.3 Vectorization: Performing math on entire columns without loops
- 4.4 Broadcasting: How NumPy handles different array shapes
- 4.5 Universal Functions (ufuncs): np.exp(), np.sqrt(), np.log()
- 4.6 Slicing and Masking: Filtering data using Boolean arrays
- 4.7 Aggregations: np.sum(), np.mean(), np.std(), np.median()
- 4.8 Reshaping and Transposing data

Unit 5 pandas — Core DataFrame Operations.....78 - 101

- 5.1 Series vs. DataFrames: The 1D and 2D structures
- 5.2 Data I/O: Reading and writing CSV, Excel, and JSON

5.3 Inspection: .head(), .info(), .describe(), .shape	
5.4 Indexing: Selecting data with .loc (labels) and .iloc (positions)	
5.5 Conditional Filtering: Selecting rows based on multiple criteria	
5.6 Column Operations: Renaming, dropping, and creating new features	
5.7 Sorting: .sort_values() and .sort_index()	
5.8 Setting and Resetting Indexes	
Unit 6 pandas — Advanced Wrangling	102 - 125
6.1 Merging: pd.merge() (Inner, Left, Right, Outer joins)	
6.2 Concatenation: pd.concat() (Stacking data vertically or horizontally)	
6.3 Groupby: Split-Apply-Combine logic	
6.4 Aggregation: .agg(['mean', 'sum', 'count'])	
6.5 Pivot Tables and Cross-tabulations (pd.pivot_table())	
6.6 Transform & Filter: Advanced Groupby operations	
6.7 Multi-indexing: Handling complex, hierarchical data	
6.8 Apply: Using custom functions on rows/columns with .apply() and .map()	
Unit 7 The Data Cleaning Protocol	126 - 149
7.1 Identifying and handling Missing Data: .isnull(), .dropna(), .fillna()	
7.2 Imputation strategies: Mean, Median, and Mode vs. Forward-fill	
7.3 Data Type Conversion: .astype() and pd.to_datetime()	
7.4 Handling Duplicates: .duplicated() and .drop_duplicates()	
7.5 String Cleaning in DataFrames: .str accessor methods	
7.6 Outlier Detection: Z-score and IQR (Interquartile Range) methods	
7.7 Regex for Data Cleaning: Using re module to fix messy patterns	
7.8 Standardizing units and category labels	
Unit 8 Data Collection (APIs & Scraping)	150 - 174
8.1 The requests library: GET and POST requests	
8.2 JSON Parsing: Converting API responses into DataFrames	
8.3 Authentication: Working with API Keys and Headers	
8.4 Web Scraping Basics: Introduction to BeautifulSoup	
8.5 Inspecting HTML: Tags, Classes, and IDs	
8.6 Scraping tables and lists into pandas	
8.7 Handling Pagination: Scraping across multiple pages	
8.8 Ethical Scraping: robots.txt and rate limiting	
Unit 9 Static Visualization (Matplotlib & Seaborn)	175 - 199
9.1 The Object-Oriented Interface: fig, ax = plt.subplots()	
9.2 Customizing Plots: Titles, Labels, Legends, and Colors	
9.3 Basic Plots: Line, Bar, Scatter, and Pie	
9.4 Statistical Viz (Seaborn): sns.histplot, sns.kdeplot, sns.boxplot	
9.5 Categorical Viz: sns.violinplot, sns.stripplot, sns.countplot	
9.6 Regression Plots: sns.regplot and sns.lmplot	
9.7 Relationship Grids: sns.pairplot and sns.heatmap	
9.8 Visualization Best Practices: Data-ink ratio and choosing the right chart	
Unit 10 Interactive Viz & Dashboards (Plotly & Streamlit)	200 - 228
10.1 Interactive Scatter and Line plots with Plotly Express	

- 10.2 Adding Interactivity: Hover data, Sliders, and Buttons
- 10.3 Geographic Maps: Choropleths and Scatter Geo
- 10.4 Streamlit Fundamentals: `st.write()`, `st.sidebar`, `st.columns`
- 10.5 Input Widgets: Selectboxes, Sliders, and File Uploaders
- 10.6 Integrating pandas with Streamlit for live filtering
- 10.7 Displaying charts and DataFrames in a web app
- 10.8 Deploying a Python Dashboard to the Cloud

Unit 11 Exploratory Data Analysis (EDA) & Stats229 - 253

- 11.1 Descriptive Statistics: Central tendency and dispersion
- 11.2 Skewness and Kurtosis: Understanding distribution shapes
- 11.3 Correlation: Pearson vs. Spearman coefficients
- 11.4 Hypothesis Testing: Null vs. Alternative hypothesis
- 11.5 T-Tests: One-sample, Independent, and Paired
- 11.6 ANOVA: Comparing means across three or more groups
- 11.7 Chi-Square Test for Independence (Categorical data)
- 11.8 P-values and Significance levels ($\alpha = 0.05$)

Unit 12 Predictive Modeling (Regression)254 - 280

- 12.1 Introduction to Scikit-Learn: The fit-predict API
- 12.2 Simple Linear Regression: OLS (Ordinary Least Squares)
- 12.3 Multiple Linear Regression: Handling multiple features
- 12.4 Assumptions of Linear Regression (Normality, Homoscedasticity)
- 12.5 Logistic Regression: Sigmoid function and binary classification
- 12.6 Model Evaluation (Regression): R^2 , MAE, MSE, RMSE
- 12.7 Model Evaluation (Classification): Confusion Matrix, Precision, Recall, F1-Score
- 12.8 Train-Test Split and Cross-Validation

Unit 13 Time Series Analysis281 - 306

- 13.1 Handling Datetime objects: `pd.to_datetime()` and `.dt` accessor
- 13.2 Resampling and Frequency Conversion (`.resample('M')`, `.asfreq()`)
- 13.3 Window Functions: Rolling means and Expanding windows
- 13.4 Time Shifting: `.shift()` and `.diff()` for growth rates
- 13.5 Seasonality and Trend decomposition
- 13.6 Handling Time Zones and Daylight Savings
- 13.7 Visualizing Time Series: Seasonal subseries plots
- 13.8 Basic Forecasting: Autocorrelation (ACF) and Moving Averages

Unit 14 Unsupervised Learning (Clustering)307 - 336

- 14.1 Introduction to Unsupervised Learning vs. Supervised
- 14.2 K-Means Clustering: How the algorithm works
- 14.3 Choosing K: The Elbow Method and Silhouette Score
- 14.4 Data Scaling: Why StandardScaler is required for clustering
- 14.5 Principal Component Analysis (PCA) for Dimensionality Reduction
- 14.6 Interpreting Cluster Centroids and Labels
- 14.7 Hierarchical Clustering (Dendrograms)
- 14.8 Real-world Case Study: Customer Segmentation